Sprint 1: User Input Testing

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Overview

My Sprint 1 task was user input. Ryan and I both wrote a version, and we decided to keep Ryan's since it was already in main. My part that went in is the tests for the mouse input and the board code, plus this document.

How the input works

- Hold the left mouse button on a tile and drag it onto a tile next to it.
- If the swap makes a match, the matched tiles clear. If it doesn't, the swap is undone.
- After a swap that makes a match, you have to let go of the mouse before you can swap again.
- Dragging onto a tile that isn't next to it does nothing, and so do the right and middle mouse buttons.

The tile under the mouse is found with `Grid.GetMousePositionInGrid`, which checks the rectangle each tile was drawn in, so it still lines up after the window is resized.

Running the tests

From the main repo folder:

```
pip install pygame pytest
python -m pytest
```

Right now this gives 27 passed and 3 xfailed. The xfailed tests are the bugs listed below. They're marked as expected failures so they don't break the test run. When one of the bugs gets fixed, its test will start failing, which means the xfail marker above that test can be removed. A warning about `pkg_resources` may also show up. That comes from pygame itself and can be ignored.

The tests are in the `tests` folder:

- `test_input.py` covers the mouse input. The game reads the mouse every frame, so the tests swap in a fake mouse and run frames of the game. No window opens.
- `test_board.py` covers the grid, swapping, matching, clearing, and refilling.
- `conftest.py` is shared setup, including a helper that fills the board from rows of letters so each test starts from a known board.

Bugs found

1. A row of 4 or more is found more than once, so `clearMatch` returns a count that's too high. The tiles still clear correctly, but this would matter if the count is used for scoring.
2. Part of an L or T shaped match can be missed.
3. Comparing a vector to None crashes instead of returning False.

Earlier versions of the tests also caught `clearMatch` crashing, `Grid.SwapTiles` not swapping, and `Vector2` multiplication rounding to whole numbers. Those have been fixed in main and now have passing tests.

Still to do

- Add tests for menu clicking once there are menus.
- Add tests for new Sprint 2 features as they get added.